

vision**2**learn

Understanding Coding

Unit 4

Learning
outcomes



Unit 4 - Understand the processes and practice of coding

Learning outcomes / Assessment criteria

1 Understand the principles of best practice in coding

- 1.1 Describe what is meant by the KISS principle
- 1.2 Give examples of the disadvantages of writing complicated code
- 1.3 Define what is meant by the single responsibility principle
- 1.4 Describe what is meant by separation of concerns
- 1.5 Define what is meant by abstraction
- 1.6 Describe what is meant by solid principles

2 Understand the methods of testing code

- 2.1 Explain what is meant by testing code
- 2.2 Explain why it is important to test code
- 2.3 Identify methods of testing code
- 2.4 Describe the characteristics of a test driven development
- 2.5 Explain the benefits of a test driven development
- 2.6 Identify what is meant by a bug in relation to code
- 2.7 Describe what is meant by debugging

3 Understand DevOps processes

3.1 Define what is meant by continuous integration

3.2 Define what is meant by continuous delivery

3.3 Define what is meant by continuous deployment

3.4 Identify the steps required for:

- continuous integration
- continuous delivery
- continuous deployment

3.5 Identify the differences between:

- continuous integration
- continuous delivery
- continuous deployment

4 Understand the importance of robust coding

4.1 Explain what is meant by exception handling

4.2 Explain what is meant by defensive programming